

Filipino Adaptation of OmniVoice for Voice Cloning and Text-to-Speech

Final Project Presentation

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The Problem

Voice cloning systems synthesize speech in a target speaker's voice from a short reference audio.

- High-quality voice cloning is strongest in high-resource languages.
- Filipino speech AI remains low-resource, with few open Filipino workflows.
- OmniVoice supports Filipino, but with only about **7.71 hours** of Filipino in its training mixture.
- The UP-DSP Philippine Languages Database holds almost **49 hours** of Filipino speech.

What We Did

Fine-tune OmniVoice on curated Filipino speech and measure whether the adapted model is better than the base model.

Background: OmniVoice

OmniVoice: Towards Omnilingual Zero-Shot Text-to-Speech with Diffusion Language Models

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Abstract

We present OmniVoice, a massively multilingual zero-shot text-to-speech (TTS) model that scales to over 600 languages. At its core is a novel diffusion language model-style discrete-continuous (NAR) architecture. Unlike conventional discrete NAR models that suffer from performance bottlenecks in complex two-stage (text-to-semantic-to-acoustic) pipelines, OmniVoice directly maps text to multi-codebook acoustic tokens. This simplified approach is facilitated by two key technical innovations: (1) a full-codebook random masking strategy for efficient training, and (2) initialization from a pre-trained LLM to ensure superior intelligibility. By leveraging a 50k-hour multilingual dataset curated entirely from open-source data, OmniVoice achieves the broadest language coverage to date and delivers state-of-the-art performance across Chinese, English, and diverse multilingual benchmarks. The code and pre-trained models are publicly available.

1 Introduction

Zero-shot text-to-speech (TTS) models trained on large-scale datasets have demonstrated a remarkable ability to generate high-quality speech conditioned on only a few seconds of reference audio [1, 2]. Despite these advances, most existing models support only a limited set of languages, often leaving hundreds of low-resource languages behind. Expanding language coverage is not merely a technical challenge but also a critical step toward creating speech technologies to languages across the globe. To address this gap, we aim to develop a massively multilingual zero-shot TTS model supporting hundreds of languages. Realizing such extensive scaling requires a TTS architecture with exceptional capacity and robustness to handle diverse linguistic patterns.

In the pursuit of such scalable and high-quality TTS, current research primarily adheres to two paradigms: autoregressive (AR) [3, 4, 5, 6, 7, 8, 9, 10, 11] and non-autoregressive (NAR) models [12, 13, 14]. NAR models offer advantages in both inference speed (via parallel decoding) and robustness (owing to bidirectional context) [12, 14, 15]. Unlike AR frameworks, models can be broadly categorized into continuous-latent based [14, 16, 17, 18] and discrete-token based [19]. The latter was shown to yield superior prosodic diversity [20].

However, state-of-the-art (SOTA) discrete-token NAR systems [19] typically rely on complex two-stage cascaded pipelines (text-to-semantic followed by semantic-to-acoustic). While such decoupling simplifies the training of individual modules, it also introduces significant drawbacks: (1) error propagation, where inaccuracies in semantic prediction degrade the final audio quality; and (2) information bottlenecks, where low-bitrate semantic representations sacrifice fine-grained acoustic details. While single-stage alternatives [21] attempt to bypass these issues, they have historically lagged behind two-stage systems in terms of speech intelligibility.

¹<https://github.com/K2-fsa/omnivoice>

Preprint. Under review.

- Open-source multilingual zero-shot TTS and voice cloning model.
- Supports 600+ languages, including Filipino (fil).
- Generates discrete acoustic tokens with a masked-token (diffusion language model) process.
- Provides official fine-tuning and evaluation scripts.
- Practical to adapt instead of training a Filipino model from scratch.

Source: Zhu et al. (2026).

Research Question

Main Research Question

How much does Filipino-specific fine-tuning improve OmniVoice for Filipino TTS and voice cloning across intelligibility, naturalness, and speaker similarity metrics?

- Model: k2-fsa/OmniVoice, adapted with supervised fine-tuning.
- Data: Filipino read speech from the UP-DSP Philippine Languages Database.
- Compute: free Kaggle T4 GPU notebooks.
- Baseline: the unmodified pretrained OmniVoice checkpoint.
- Tracking: all runs logged in Weights and Biases.

Dataset: UP-DSP Philippine Languages Database



The screenshot shows the Mozilla Data Collective interface for the 'UP - DSP - Philippine Languages Database (UP-DSP-PLD)'. The page header includes the Mozilla Data Collective logo and navigation links: 'Browse Datasets', 'API', 'Contribute Datasets', and 'Data Assistant'. The main title is 'UP - DSP - Philippine Languages Database (UP-DSP-PLD)'. Below the title, it specifies the license as 'CC-BY-NC-4.0' and the steward as 'UP-EEEL - Digital Signal...'. Metadata includes 'Task: ASR', 'Release Date: 3/19/2026', 'Format: WAV, LOG', and 'Size: 45.63 GB'. There are buttons for 'Download' and 'Connect API', and a 'Share' icon. The 'Description' section states: 'This dataset contains multilingual, text and speech pairs for ten Philippine languages namely Filipino, English, ...'.

- Released through Mozilla Data Collective (CC-BY-NC-4.0).
- WAV audio with LOG transcript files.
- Filipino subset: 135 speakers, 52,879 utterances, 48:56:36.
- We used read speech only: the transcript reliably matches the audio.

Source: Cajote et al. (2024).

Data Preparation

Conservative filtering: read speech only, no transcripts with digits or parentheses, no broken WAVs, clips between 1 and 15 seconds, speaker-disjoint splits.

Split	Samples	Percent	Hours	Speakers
Train	33,448	79.1%	31.9	103
Development	4,506	10.7%	3.7	13
Test	4,322	10.2%	4.2	13
Total	42,276	100%	39.8	129

The 31.9 training hours are about $4\times$ the Filipino hours in OmniVoice's original training mixture. Domains include news, medical sentences, stories, greetings, proverbs, and many isolated-word prompts.

System Architecture

1. Dataset preparation

PLD Filipino
WAV audio +
LOG transcripts

Cleaning
read speech only
1.0s-15.0s

Speaker-disjoint
train / dev / test
JSONL splits

2. Tokenization and supervised fine-tuning

8-codebook acoustic tokens

text

Higgs Audio v2
continuous waveform
to discrete tokens

8
layers



masked
target

OmniVoice
pretrained mul-
tilingual TTS
masked diffusion LM

SFT objective
weighted
cross-entropy

Best-dev
checkpoint
selection
LR comparison

3. Held-out evaluation and delivery

Zero-shot inference
matched test
prompts
+ reference voices

Generated
Filipino speech

Whisper ASR
WER

ECAPA-TDNN
SIM-o

UTMOS
naturalness

React audio
showcase

Cloudflare
Workers

selected checkpoint

Fine-Tuning Setup

- Environment: Kaggle notebooks, $2 \times$ NVIDIA T4, FP16, SDPA attention, gradient accumulation.
- Start from the pretrained k2-fsa/OmniVoice checkpoint.
- Loss: OmniVoice's own objective, weighted cross-entropy on masked acoustic tokens.
- Optimizer: AdamW, cosine learning-rate decay, 1% warmup, seed 42.

Run	Learning rate	Steps	Checkpoint policy
FT-5e-6	5e-6	5000	lowest dev loss (found at step 5000)
FT-1e-5	1e-5	5000	lowest dev loss (found at step 4900)
FT-2e-5	2e-5	5000	lowest dev loss (found at step 5000)

A **controlled comparison**: same steps, same checkpoint policy, only the learning rate differs.

Training Behavior



W&B training-loss curve for the 5000-step LR $1e-5$ run, whose best-development-loss checkpoint (step 4900) is the recommended model.

How We Evaluated

All systems generate the same 4,322 held-out test utterances, cloning a reference voice from the same speaker (a different utterance, so the model never hears the target audio).

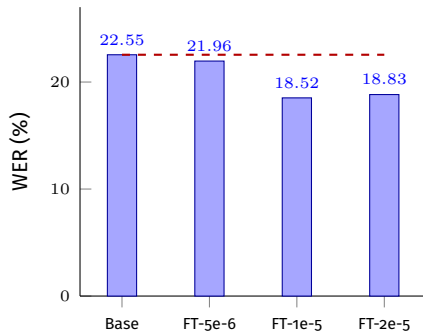
- **WER: “Did it say the right words?”**
Whisper-large-v3 transcribes the generated audio; transcript vs. target text. Lower is better.
- **SIM-o: “Does it sound like the reference speaker?”**
Cosine similarity of speaker embeddings (WavLM ECAPA-TDNN). Higher is better.
- **UTMOS: “Does it sound natural?”**
A learned mean-opinion-score predictor. Higher is better.

Same test split, prompts, inference settings, and evaluator models for every system, so differences come from the checkpoints.

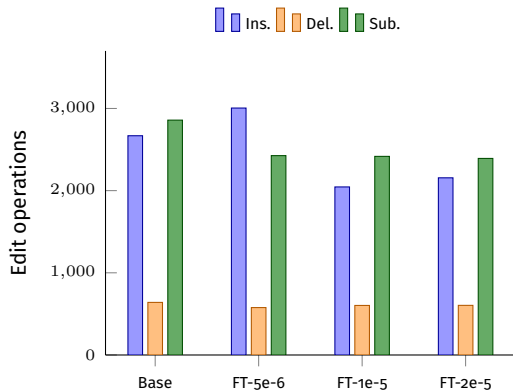
Results: Main Comparison

Model	WER (%)	SIM-o	UTMOS
Base	22.55	0.602	3.64
FT-5e-6	21.96	0.605	3.60
FT-1e-5	18.52	0.604	3.61
FT-2e-5	18.83	0.583	3.61

FT-1e-5 cuts WER by **4.03 points** (17.9% relative) while keeping SIM-o above base. FT-2e-5 is close in WER but loses speaker similarity.



Results: Where the Improvement Comes From



- FT-1e-5 wins by producing **fewer extra words** (insertions: 2,668 $\rightarrow$ 2,045) and **fewer wrong words** (substitutions: 2,858 $\rightarrow$ 2,417).
- FT-2e-5 shows the same pattern, slightly behind on insertions.
- FT-5e-6 cut substitutions but its insertions rose to 3,005, absorbing most of the gain.
- A run can improve one error type while worsening another.

Discussion

- The learning rate controls a **tradeoff**: 1e-5 and 2e-5 buy intelligibility; 5e-6 stays closest to the base model's speaker similarity.
- FT-2e-5 shows the cost of pushing harder: nearly the same WER as FT-1e-5, but the **only SIM-o below base** (0.583).
- The base model keeps the highest UTMOS (3.64); fine-tunes are close at 3.60–3.61 — a small, consistent naturalness cost.
- No single metric picks the winner; FT-1e-5 wins on the **overall pattern**: nearly all the WER gain with SIM-o preserved.
- Practical lessons: tokenize once and reuse; a conservative FP16 + gradient-accumulation recipe makes T4 training feasible.

Limitations

- Automatic metrics only: WER is ASR-mediated (Whisper), SIM-o depends on the speaker encoder, UTMOS is a predictor, not a listening survey.
- Read speech only, with many short isolated-word prompts; says little about spontaneous speech or long-form prosody.
- Filipino only; other Philippine languages in PLD were not tested.
- Three learning-rate settings, not a full sweep, because of free-GPU compute limits.

Conclusion and Recommendations

Answer to the Research Question

Filipino fine-tuning improved OmniVoice intelligibility: WER dropped from 22.55% to 18.52% (17.9% relative) with speaker similarity preserved. The learning rate controls an intelligibility-versus-speaker-similarity tradeoff; no setting wins every metric.

- Recommended checkpoint: **FT-1e-5** (best dev loss, found at step 4900).
- Next: Filipino listener study to validate the automatic metrics.
- Next: more learning rates around $1e-5$; repeat runs across seeds.
- Next: curated sentence-level data to improve prosody.

Application Use Case: Voice Cloning

 **OmniVoice · Filipino Fine-tuning**
Can an AI clone a Filipino voice from one short recording? Compare the base model against three fine-tuned variants.

 dark mode

TEST UTTERANCES

01
spk_0105
Pero kampante naman ang EU na makakahanap ng agarang solusyon an...

02
spk_0085
Nuong panahon nang ang Pilipinas ay nasasakop pa ng mga Kastila ay...

03
spk_0093
Nakahanda ang bansa na kunin ang malaking parte ng pamilihang-mangga...

04
spk_0153
Naniniwala ang mga mangangalakal na lalong lalago ang negosyo sa lungsod ...

05
spk_0166
Kung gayon, dapat ay gawin matn ang ating tungkulin na mas ipasalam ang...

06
spk_0002
diko

07
spk_0002
bunso

08
spk_0002
amang

09
spk_0002
kastambal

1 Models hear a voice prompt — a short clip of the speaker saying a different line — the only audio they get.

2 Models must say the target text — in that speaker's voice, without ever hearing the real recording of it.

3 You compare the attempts — each AI clone next to the ground truth — the real speaker reading the same line.

Sample 01 Filipino speaker 0105 0105.111124.050515.0394

TARGET TEXT

"Pero kampante naman ang EU na makakahanap ng agarang solusyon ang Pilipinas patungkol sa naturang isyu."

Voice prompt steps 1-2 · input

The only audio the models heard: speaker 0105 saying a *different* line.



"Nakahanda ang bansa na kunin ang malaking parte ng pamilihang-mangga sa Estados Unidos, ayun sa sekretaryo ng Departamento ng Agrikultura."

Listen & compare step 3 · output

Ground truth is the real recording; the rest are AI clones. Switch while playing — the position is kept, so you hear the exact same moment in a different voice.

Ground truth 1 Base 2 LR 2e-5 3 LR 5e-6 4 LR 1e-5 5



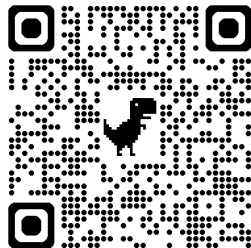
Waveform visualization showing audio amplitude over time. The interface is dark-themed with purple accents.

Public Showcase

OmniVoice Filipino Showcase

Public demo for comparing reference audio, base OmniVoice, and Filipino fine-tuned outputs from the evaluation samples.

<https://omnivoice-showcase.duncanb013.workers.dev/>



References

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- Weights & Biases experiment report:
<https://wandb.ai/duncanb013-polytechnic-university-of-the-philippines/OmniVoice-PLD-Filipino>